S-BLOCK ELEMENTS

ALKALI METALS

Element	Atomic no.	Atomic mass	Flame colour
Li (Lithium)	3	6.941	Li/Li^+ – Crimson Red
Na (Sodium)	11	22.990	Na/Na^+ – Golden Yellow
K (Potassium)	19	39.098	K/K^+ – Violet
Rb (Rubidium)	37	85.468	Rb/Rb^+ – Red Violet
Cs (Caesium)	55	132.905	Cs/Cs^+ – Blue
Fr (Francium)	87	223.020	

ATOMIC SIZE
Li < Na < K < Rb < Cs
$Li^+ < Na^+ < K^+ < Rb^+ < Cs^+$

IONISATION ENERGY
Li > Na > K > Rb > Cs

ELECTROPOSITIVE CHARACTER / REACTIVITY
$\propto \frac{1}{IE}$
Li < Na < K < Rb < Cs

MELTING POINT
Li > Na > K > Rb > Cs

BOILING POINT
Li > Na > K > Rb > Cs

HYDRATION ENERGY
$Li^+ > Na^+ > K^+ > Rb^+ > Cs^+$

COMPLEX FORMATION
In alkali metals, only Li form complex due to small size
eg: $LiCl.2H_2O$

IONIC MOBILITY
$\left(\propto \frac{1}{\text{hydration}}\right)$
$Li^+ < Na^+ < K^+ < Rb^+ < Cs^+$

DENSITY
Li < Na < K < Rb < Cs

METAL HYDRIDES
$2M + H_2 \rightarrow 2MH$
LiH, NaH, KH, RbH, CsH
- **THERMAL STABILITY**
LiH > NaH > KH > RbH > CsH
- **REDUCING CHARACTER**
LiH < NaH < KH < RbH < CsH

REACTION WITH OXYGEN
Li forms only normal oxide Li_2O
Na forms normal oxide (Na_2O) & Peroxide (Na_2O_2)
K forms normal oxide, Peroxide & Superoxide (KO_2)
Rb forms normal oxide, Peroxide & Superoxide
Cs forms normal oxide, Peroxide & Superoxide
KO_2, RbO_2, CsO_2 (Superoxide) - Paramagnetic & coloured

METAL HYDROXIDES
BASIC CHARACTER
LiOH < NaOH < KOH < RbOH < CsOH

REACTION WITH H_2O
Li React with water
Na, K React easily with water (fast reaction)
Rb, Cs Give explosive reaction with water

REACTION WITH NITROGEN
Only Li react with N_2 gas to form Li_3N which on hydrolysis give NH_3 gas.
$6Li_{(s)} + N_{2(g)} \rightarrow 2Li_3N$
Na/K/Rb/Cs + N_2 → no reaction
$Li_3N + 3H_2O \rightarrow 3LiOH + NH_3$

COMPOUNDS OF SODIUM

Sodium carbonate/ washing soda
$Na_2CO_3 . 10H_2O$
Solvay Process:
Reactants- NH_3, CO_2
Regeneration of NH_3 - Using $Ca(OH)_2$
Byproduct - $CaCl_2$
we can't obtain K_2CO_3 Because $KHCO_3$ obtained in 2nd step is water soluble

Caustic Soda NaOH
(i) It is a white crystalline solid
(ii) Deliquescent in nature
(iii) Reaction with acidic oxide
$NaOH + CO_2 \rightarrow Na_2CO_3 + H_2O$

Sodium bicarbonate/ Baking Soda/ $NaHCO_3$
(i) It is white crystalline solid
(ii) It is sparingly soluble in water
(iii) It is not stored with strong bases like NaOH, because it has acidic H
$NaHCO_3 + NaOH \rightarrow Na_2CO_3 + H_2O$

Anomalous behaviour of Li (Due to small size & High charge density)
- The melting point and boiling point of lithium are higher that than other alkali metals.
- The hardness of lithium is higher than other metals.
- The alkali metal chlorides do not have the capability to form hydrates but lithium chloride crystallizes to form a hydrate
- Lithium nitrate decomposes to form an oxide whereas other metals on heating give nitrites.
- Compounds of lithium are partially soluble in water whereas the alkali metals are highly soluble in water.

Diagonal relationship with Mg
- Same $\frac{\text{Charge}}{\text{Radius}}$
- Li & Mg → harder
- Li & Mg react with cold water
- Li & Mg both form nitride which on hydrolysis give NH_3 gas
$Li_3N + 3H_2O \rightarrow 3LiOH + NH_3$
$Mg_3N_2 + 6H_2O \rightarrow 3Mg(OH)_2 + 2NH_3$
- LiCl and $MgCl_2$ both are deliquescent in nature.

ALKALINE EARTH METALS

Element	Atomic no.	Flame colour
Be	4	
Mg	12	
Ca	20	Ca/Ca^{2+} – Brick Red
Sr	38	Sr/Sr^{2+} – Crimson
Ba	56	Ba/Ba^{2+} – Apple Green
Ra	88	

ATOMIC SIZE
Be < Mg < Ca < Sr < Ba

IONISATION ENERGY
Be > Mg > Ca > Sr > Ba

ELECTROPOSITIVE CHARACTER / REACTIVITY
$\left(\propto \frac{1}{IE}\right)$
Be < Mg < Ca < Sr < Ba

MELTING POINT
Be > Ca > Sr > Ba > Mg

BOILING POINT
Be > Ba > Ca > Sr > Mg

HYDRATION ENERGY
$Be^{2+} > Mg^{2+} > Ca^{2+} > Sr^{2+} > Ba^{2+}$

COMPLEX FORMATION
Be, Mg & Ca form complexes
- $BeCl_2 + 2H_2O \rightarrow BeCl_2 . 2H_2O$
- $BeF_2 + 2F^- \rightarrow BeF_4^{2-}$ (max. C.N = 4)
- Ca+EDTA → $[Ca(EDTA)]^{2-}$ (C.N = 6)

IONIC MOBILITY
$\propto \frac{1}{\text{hydration}}$
$Be^{2+} < Mg^{2+} < Ca^{2+} < Sr^{2+} < Ba^{2+}$

DENSITY
Ca < Mg < Be < Sr < Ba

METAL HYDRIDES
MgH_2, CaH_2, SrH_2, BaH_2
- **THERMAL STABILITY**
$BeH_2 > MgH_2 > CaH_2 > SrH_2 > BaH_2$
- **REDUCING CHARACTER**
$BeH_2 < MgH_2 < CaH_2 < SrH_2 < BaH_2$

REACTION WITH OXYGEN
1) Be, Mg → Normal oxide BeO & MgO
2) Ca, Sr, Ba → Normal oxide & Peroxide. Peroxide is diamagnetic, but shows colour due to Lattice defect

METAL HYDROXIDES
BASIC CHARACTER
$Mg(OH)_2 < Ca(OH)_2 < Sr(OH)_2 < Ba(OH)_2$

REACTION WITH H_2O
1) Be → Steam
2) Mg → Hot water
3) Ca, Sr, Ba → React with cold water

REACTION WITH NITROGEN
All elements react with nitrogen to form nitrides of type M_3N_2
$Mg_3N_2 + H_2O \rightarrow Mg(OH)_2 + 2NH_3$

COMPOUNDS OF CALCIUM

Calcium oxide/ Quick lime/ CaO
1) $CaO + H_2O \rightarrow Ca(OH)_2$ (Quick lime → Slaked lime)
Uses
Used as basic flux
$CaO + SiO_2 \rightarrow CaSiO_3$ (basic flux + acidic impurity → slag)

Slaked lime/$Ca(OH)_2$
1) Used in white wash
2) In manufacturing of bleaching powder, glass etc
$Ca(OH)_2 + Cl_2 \rightarrow CaOCl_2 + H_2O$ (Bleaching powder)
3) In removing temporary hardness of water

$CaSO_4.2H_2O$ Gypsum
↓
$CaSO_4.1/2H_2O$ Plaster of Paris
↓
$CaSO_4$ Dead burnt plaster

Anomalous behaviour of Be (Due to small size & High charge density)
1) The coordinate number of Be is not more than 4 where as other alkali metals have coordination number of 6
2) The M.P and B.P are higher when compared to the other elements of the group
3) They forms covalent bonds whereas the other members of the group forms ionic bonds
4) Be does not react with H_2O like the other companions of the group

Diagonal relationship with Al
- Same $\frac{\text{Charge}}{\text{Radius}}$
- Be & Al are not attacked easily by acids[HNO_3] forms protective layer
- $BeCl_2$ & $AlCl_3$ → Lewis acid and form dimer
- $BeCl_2$ in vapour as dimer and solid as polymer
- BeO, $Be(OH)_2$, Al_2O_3, $Al(OH)_3$ → Amphoteric
- Both form complexes - $[BeF_4]^{2-}$–sp^3 $[Al(H_2O)_6]^{3+}$–sp^3d^2

PHYSICS WALLAH